Jedidiah Alindogan

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EDUCATION

Ph.D. in Control and Dynamical Systems
California Institute of Technology
09/2024 - Current

B.S. in Mechanical Engineering
California Institute of Technology
09/2021 - 06/2023

A.S. in - Physics, Math, Mechanical Engineering, and Electrical Engineering
American River College
06/2018 - 06/2021

SKILLS

Programming Languages
- Python, C++, C#

ML/Robotics
- PyTorch, MuJoCo, PPO, ROS 1, ROS 2, Gazebo

Computation
- MATLAB, Mathematica

Design
- Solidworks, Fusion360, ANSYS

LINKS

jedi-alindogan.github.io
github.com/jedi-alindogan
www.linkedin.com/in/495-jedidiah-alindogan

HONORS & AWARDS

2022 - Tau Beta Pi Honors Society Member

2021 - 2022 NASA BIG Idea Challenge Finalist

2019 - NASA Community College Aerospace Scholar

RELEVANT COURSES

Manipulation, Kinematics, Dynamics;
Planning, Navigation, and Perception;
Robotics Lab: Localization and Perception;

Optimization;
Linear Functional Analysis;
Probability and Measure Theory;
Probability Models;

Data-Driven Control;
Nonlinear Dynamics and Control;
Linear System Control;
Optimal Control and Estimation;

EXPERIENCE

Graduate Student - Caltech 09/2024 - Present
- *RL Locomotion for Snake Robot (Caltech–NASA JPL)*
  - Python, MuJoCo, Proximal Policy Optimization (PPO)
  - Developed a PPO-based planar locomotion controller for an 18-DOF snake robot, using structured sinusoidal gait actions to track commanded linear and angular velocities.
  - Designed a unicycle-integrated pose-tracking reward to mitigate oscillatory velocity feedback; structured gait constraints improved training convergence by up to 40%.
  - Validated zero-shot sim-to-real transfer: policies trained entirely in MuJoCo operated on concrete and gravel during building-foundation inspection trials.

Research Engineer - Caltech 08/2023 - 08/2024
Experimentation, Simulation, Python, ROS, Gazebo; Actively contribute to the design and execution of experiments, employ advanced simulations, and engage in software development within the Autonomous Robotics and Control Lab
- *Autonomous Offroad Sensing Platform*
  - C++, Python, Arduino
  - Developed autonomous payload for a Traxxas Xmaxx to run planning and control algorithms. This included design of a payload container through Solidworks, wiring of the sensors and motors to the Jetson Orin, and calibration of the sensors and motors. Calibrating the motors was done through ODrive's online GUI, and visual-inertial odometry sensor calibration was accomplished via online packages such as OALiCalib and DLIO.
- *Learning Introspective Control*
  - C++, Python, Gazebo, Git
  - Created a tipping safety constraint, conducted experiments for control and planning algorithms, designed simulation terrains for testing in Gazebo. The tipping safety constraint used physics-based estimation of centrifugal forces when turning on slants to prevent tipping. For experimentation, I validated neural-network based controllers on the robot over different terrain.

PAPERS

"Vertical Mobility and RL-Based Locomotion of a Snake-Like Robot for Building Foundation Inspection." Michael Paton, Jedidiah D. Alindogan, Shuhei Shigeto, Richard Rieber, et al. *IEEE/RSJ IROS*, 2026.

"MAGIC^{VFM}: Meta-learning Adaptation for Ground Interaction Control with Visual Foundation Models." Elena Sorina Lupu, Fengze Xie, James A. Preiss, Jedidiah Alindogan, Matthew Anderson, Soon-Jo Chung. *IEEE Transactions on Robotics*, 2025.

"Model Predictive Trees: Sample-Efficient Receding Horizon Planning with Reusable Tree Search." John Lathrop, Benjamin Rivière, Jedidiah Alindogan, Soon-Jo Chung. Presented at 2024 IEEE/RSJ.

"Lunar Architecture for Tree-Traversal In-service-of Cabled Exploration (LATTICE)." Technical paper submitted to 2022 NASA BIG Idea.

VOLUNTEERING

Academics Chair - Graduate Student Council 2024-Present
- Coordinate activities aimed at enhancing academic and career interests of graduate students and to help foster an environment in which graduate students can learn to become leaders in their academic fields.

Rise Tutor - Caltech Y 2022-2023
- Advise, mentor, and tutor high school students in STEM subjects.

Social Director Team - Avery House, Caltech 2022-2023
- Volunteer in organizing, planning, and running social events for student life on campus.

Fair Oaks Library Volunteer - Fair Oaks Public Library 2018-2021
- More than 120 hours volunteering at the Fair Oaks Public Library to help with community events and organize texts.